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Terrestriality across the primate order: A review and analysis of ground use in primates

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Abstract

Terrestriality is relatively rare in the predominantly arboreal primate order. How frequently, and when, terrestriality appears in primate evolution, and the factors that influence this behavior, are not well understood. To investigate this, we compiled data describing terrestriality in 515 extant nonhuman primate taxa. We describe the geographic and phylogenetic distribution of terrestriality, including an ancestral state reconstruction estimating the frequency and timing of evolutionary transitions to terrestriality. We review hypotheses concerning the evolution of primate terrestriality and test these using data we collected pertaining to characteristics including body mass and diet, and ecological factors including forest structure, food availability, weather, and predation pressure. Using Bayesian analyses, we find body mass and normalized difference vegetation index are the most reliable predictors of terrestriality. When considering subsets of taxa, we find ecological factors such as forest height and rainfall, and not body mass, are the most reliable predictors of terrestriality for platyrrhines and lemurs.

KEYWORDS

ancestral reconstruction, evolution of terrestriality, ground use, primate arboreality, primate ecology, primate terrestriality

1 | INTRODUCTION

Among all vertebrates, the living primates show the widest range of locomotor adaptations to arboreality.[1] Arboreality is the primitive condition in the primate order,[2] having initially evolved in the ancestral mammalian grandorder Euarchonta. A subsequent evolutionary shift to a ground-dwelling lifestyle (terrestriality) occurred repeatedly across the order and is most apparent in the African apes, African and Asian monkeys, and humans,[3] many of which evolved morphological (e.g., reduction or loss of the tail[4]), locomotor (e.g., knuckle-walking[5]), and dietary (e.g., graminivory[6]) adaptations to a terrestrial niche. However, the evolutionary factors that influenced the transition of some primate lineages from an arboreal lifestyle to a terrestrial one are still not well understood.

Here, we review terrestriality across the primate order, detailing its distribution geographically and phylogenetically, including a reconstruction of the possible evolution of terrestriality in nonhuman primates. Then, we consider popular hypotheses concerning the primate transition from arboreality to terrestriality, weighing the evidence for each. We present an analysis of existing data concerning extant primate terrestrial behavior and consider which ecological and physiological factors best predict contemporary primate ground use and if this evidence supports predictions based on current hypotheses. We close with suggestions for future research.

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2 | THE DISTRIBUTION OF TERRESTRIALITY IN PRIMATES

Understanding the extent of terrestriality among primates is crucial for unraveling their ecological adaptations and evolutionary history. Describing where and which primates are terrestrial is an important step in assessing the prevalence of this behavior across the primate order and allows us to infer when and how many times terrestriality arose throughout primate evolution. It is likely that terrestriality evolved multiple times, evident in the extant terrestrial strepsirrhines, hominins, and cercopithecines.[7] How many times this has happened, however, remains unclear.[4] While previous work has examined the evolution of terrestriality in certain clades, finding chromosomal evidence for a single transition to terrestriality in guenons, for example,[8] an analysis of the evolution of this trait across the entire primate order is still needed. Using published data pertaining to terrestriality in primates we first explore the geographic and phylogenetic distribution of terrestriality in the primate order, and then use ancestral state reconstructions (ASR) to estimate when and how often terrestriality evolved.

2.1 | Terrestriality data

We collated primate terrestriality data from published sources including online databases and journal articles. Databases included All The World's Primates (ATWP; Rowe & Myers[9]), Animal Diversity Web (ADW[10]), and PanTHERIA.[11] Publications comparing large numbers of primates were particularly useful (e.g., DeCasien and colleagues,[12–14]). We conducted literature searches using Web of Science and Google Scholar using keywords SPECIES NAME plus ARBOREAL, TERRESTRIAL, GROUND USE, STRATA USE, and/or HABITAT USE. We focused primarily on collecting terrestriality data at the species level, considering subspecies only for the apes and when data were readily available. To consider as many taxa as possible, we excluded species from counts and subsequent analyses only when terrestriality data for all associated subspecies were available. We conducted literature searches between October 18, 2021 and January 26, 2022, following the taxonomy of the IUCN which listed 787 extant primate taxa during that period.

Existing data describe primate terrestrial behavior using a binary (terrestrial or arboreal), ordinal (terrestrial, semiterrestrial, or arboreal), or continuous (percentage of active time spent on the ground) classification. We designated the terrestriality of taxa using each classification system.

We classified 515 primate taxa using the binary classification, using either existing binary designations or classifying species based on percentage of ground use. Modifying the methods used by Arenson et al.,[15] we classified a taxon as arboreal if it used the ground up to 40% of the time and terrestrial if ground use was greater than 40%. When binary or continuous terrestriality data were not available, we classified taxa based on descriptions in the literature of a taxon being "strictly" or "highly" arboreal or terrestrial. Agreement between databases and publications for binary terrestriality data was high. For example, binary data for 219 taxa were available from two or more sources, and for 203 of these taxa (93%) the classification of arboreal or terrestrial was identical between sources. Of the 16 taxa that had multiple data sources that were not in agreement (i.e., a mix of "arboreal" and "terrestrial" classifications), we classified 10 as semiterrestrial, and four were arboreal primates that show a moderate amount (>10%) of ground use.

We classified 426 primate taxa using an ordinal classification of terrestriality. We created an ordinal classification for each taxon for which data on percentage ground use was available. Following Arenson et al.,[15] we classified taxa that used the ground less than 20% of the time as arboreal, between 20% and 40% as semiterrestrial, and more than 40% as terrestrial. When continuous terrestriality data were not available, we used descriptions of taxa being "strictly" or "highly" arboreal or terrestrial to classify them as arboreal or terrestrial, and descriptions of "semiarboreal" or "semiterrestrial" to classify taxa as semiterrestrial.

We classified 239 primate taxa using a continuous classification of terrestriality, based on the reported average percent of active time spent on the ground. Published estimates of primate ground use range from 0% to 98%. The distribution is highly left-skewed, indicating most primates are highly arboreal (median 1%, interquartile range 0%–8%). Of the 239 taxa that have continuous data, 27 (~11%) had data from both ATWP and from the literature. In each case, we opted to use terrestriality estimates from the latter.

2.2 | Terrestriality mostly in Africa and Asia

Of the taxa for which there are continuous terrestriality data, 228 also have spatial data and are included in the map in Figure 1. Most terrestrial primates are found in Africa and Asia (Figure 1b,c). While there are no highly terrestrial primates in Central or South America, three species show moderate levels of terrestriality; *Sapajus libidinosus* (27% time on the ground), *Sapajus flavius* (25%), and *Cebus olivaceus* (18%; Figure 1a). Terrestrial primates are mostly in the tropics, although some are in subtropical regions as well. In fact, numerous terrestrial primates range north of the Tropic of Cancer (23° N), including *Macaca arctoides*, *Macaca fuscata*, *Macaca mulatta*, *Macaca sylvanus*, *Macaca thibetana*, and *Semnopithecus ajax*.

2.3 | Terrestriality is concentrated within Papionini

In Figure 2, we detail the phylogenetic distribution of terrestriality among extant primates, using the ordinal classification of terrestriality and the consensus primate phylogenetic tree from 10kTrees, ver.c 3, using the GenBank taxonomy.[16] Most taxa classified as terrestrial or semiterrestrial are in the super-family Cercopithecoidea, especially tribe Papionini. This clade includes the most terrestrial primates, principally in *Macaca*, *Mandrillus*, *Papio*, and *Theropithecus*.

Outside the cercopithecoids, terrestrial taxa are found exclusively in Hominoidea, specifically the African apes. Two species of lemurs are classified as semiterrestrial as well, completing our list of taxa that use the

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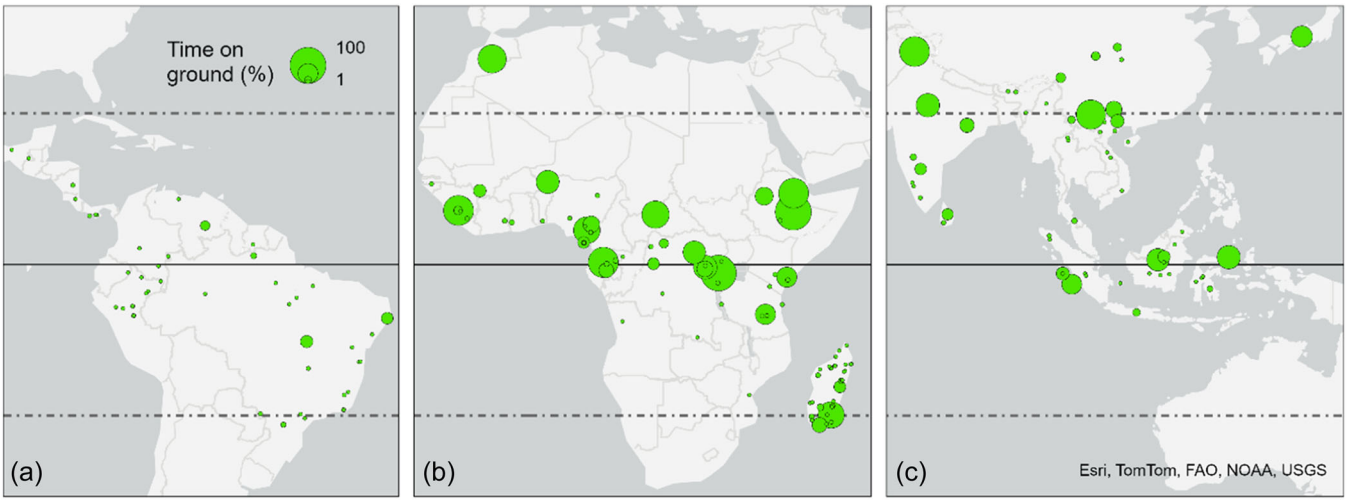

FIGURE 1 Geographic distribution of primate terrestriality across the Americas (a), Africa (b), and Asia (c) using the continuous terrestriality data. Points represent primate range centroids and are proportional in size to time spent on the ground—legend depicts sizes for 100%, 50%, and 1% ground use. Horizontal solid line depicts the equator, dashed lines delineate the tropics.

ground to an appreciable degree. Some taxa with substantial degrees of terrestriality are not visible in this phylogenetic tree because they reportedly spend less than 20% of the time on the ground and are therefore excluded from our "semiterrestrial" category. These taxa are in the genera *Lemur*, *Rhinopithecus*, *Semnopithecus*, and *Trachypithecus*, and are visible in the phylogenetic tree constructed using continuous data (Supporting Information S1: Figure S1).

2.4 | Ancestral primate terrestriality

To estimate when and how many times terrestriality evolved in primates, we conducted ASR using all three terrestriality classifications. A strong phylogenetic signal using Pagel's lambda (λ) was found for each classification: binary (λ = 0.89, maximum-likelihood estimate [MLE] = −75.8), ordinal (λ = 0.86, MLE = −111.9), and continuous (λ = 0.73, MLE = −597.4), indicating the appropriateness of an ancestral reconstruction of the evolutionary pathways that resulted in the observed terrestriality of extant primates. We used the *ape* package in R[17] to conduct the ASR using the 10kTrees phylogeny. The results of the ASR using the ordinal terrestriality data, showing the posterior probabilities of ancestral taxa being arboreal, semi-terrestrial, or terrestrial, are shown in Supporting Information S1: Figure S1. The results of the ASR using the continuous data are shown in Supporting Information S1: Figure S2. As the phylogenetic distribution using binary data was highly similar to that produced using ordinal data, we do not present ASR using the binary data here.

Based on the ordinal ASR results, it is plausible that terrestriality evolved on as few as two occasions throughout primate evolution: specifically, within the Homininae and Cercopithecinae lineages, during the Miocene epoch, around 15–9 Ma and 22–15 Ma, respectively. However, the ASR using continuous data suggests a more nuanced history of primate terrestriality, with considerable terrestrial behavior apparent in more lineages than appear in the ordinal ASR, specifically among Asian colobine genera including *Rhinopithecus*, *Semnopithecus*, and *Trachypithecus*, and Malagasy species including *Daubentonia madagascariensis* and *Lemur catta*.

It is notable that the possible timing of the transitions to terrestriality indicated here coincide with the major global climatic changes characteristic of the Miocene epoch, including a prolonged cooling event, the expansion of grasslands, and recession of forests. These significant ecological shifts likely exerted strong selection pressures on primates, potentially prompting the exploitation of terrestrial niches within specific clades.

3 | THEORIES OF TERRESTRIALITY

Multiple hypotheses have been posited to explain why nonhuman primates, all of which navigate an arboreal space to some degree, engage in terrestrial behaviors. Here we consider five:

1. Body mass is positively associated with terrestriality—heavier, larger-bodied primates are more likely to be terrestrial than lighter, smaller species.
2. Forest complexity is negatively associated with terrestriality—primates in more structurally complex forests are less likely to be terrestrial.
3. Predation pressure is negatively associated with terrestriality—primates are less likely to be terrestrial in habitats that have many terrestrial predators.
4. Arboreal food abundance is negatively associated with terrestriality—primates in habitats with abundant arboreal food resources are less likely to be terrestrial.
5. Weather variables are positively associated with terrestriality—primates that inhabit warmer and/or drier habitats are more likely to be terrestrial.

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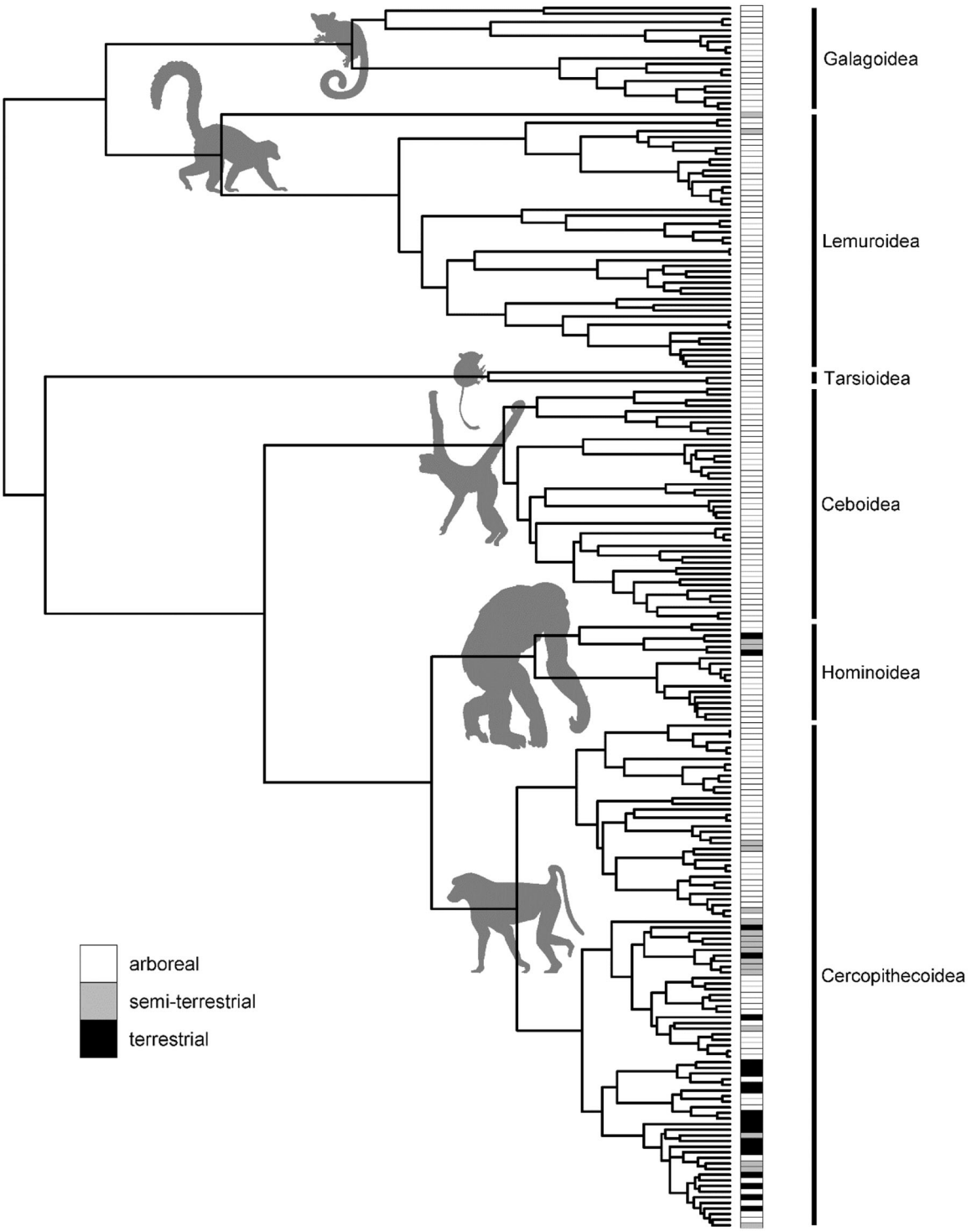

FIGURE 2 Phylogenetic distribution of terrestriality using the ordinal terrestriality classification.

3.1 | Body mass

Body size is one of the most ecologically significant attributes of any animal[18] and possibly the single most useful predictor of species' adaptations in mammals.[19] Body mass has been found to be a key indicator of mammalian characteristics, such as basal metabolic rate,[18] total energy expenditure,[20] activity period (e.g., nocturnal or diurnal[21]), and mode of locomotion.[22] Primate substrate use has also traditionally been viewed as a consequence of body mass, with larger primates purportedly more likely to use the ground than smaller primates (e.g., Clutton-Brock & Harvey[23]).

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One argument for the influence of body mass on primate terrestrial behavior is an increase in locomotor energetic efficiency. Body mass influences locomotor costs and has implications for terrestrial and arboreal movement.[24–26] Models of primate socioecology and behavioral ecology assume high energetic costs of travel (e.g., Chapman & Chapman[27]) but physiological studies suggest that for both arboreal and terrestrial species distance traveled each day accounts for only 2%–10% of daily energy expenditure.[28,29] This suggests that the energetic costs associated with locomotion, and thereby overall body mass, may not be as strong predictors of primate locomotor behavior as has been previously assumed.[30] The precise ways in which locomotion imposes energetic costs, however, are likely to differ as a function of body size. For example, while it has been argued that long arboreal travel bouts are more energetically expensive for primates than descending to the forest floor, traveling terrestrially, and reascending,[31] this may not apply to primates of all sizes. The energetic cost of vertical climbing was found to be significantly higher than the cost of horizontal locomotion for large primates (>0.5 kg) but not for smaller primates (<0.5 kg[32]). Further, large body sizes have been argued to be more energetically efficient for long-distance terrestrial travel, evidenced by the more terrestrial cercopithecines being considerably larger than the more arboreal cercopithecines.[5]

3.2 | Forest characteristics

An early argument for the transition from arboreality to terrestriality was the widespread change in primate habitats.[33] The arboreal nature of archaic catarrhines, for example, has been associated with inhabiting dense forest habitats, while the terrestrial adaptations of the cercopithecoids have been associated with inhabiting more open savannah habitats.[3]

Primate presence, population densities, and species richness have been associated with aspects of forest structure, such as forest canopy stratification,[34] the presence of tall trees,[35] and canopy cover.[36] Recent research has focused on determining the influence of forest structure on primate behavior as well. For example, different forest structural characteristics such as forest height and canopy stratification were found to influence the sleeping, ranging, and feeding activities of two sympatric primates in different ways.[37] Forest structure has also been associated with increased terrestrial behavior in primates. Capuchins living in a tropical dry forest in Eastern Brazil, for example, were found to spend a comparatively large amount of time (up to 44%) on the ground.[38] Similarly, canopy connectivity can influence primate ground use. For example, Golden snub-nosed monkeys frequently descend to the ground when faced with large gaps in the canopy.[39]

3.3 | Predation pressure

An increase in terrestrial behaviors has long been argued to be associated with an increase in predation risk for primates. Some primates utilize distinct alarm calls in response to terrestrial and airborne threats.[40] Also, it has been suggested that some primates can assess the relative threat of different terrestrial predators such as different species of snakes.[41] Further, some arboreal species increase the amount of time they spend on the ground only when engaging in polyspecific associations with other primates that act as sentinels for terrestrial predators, decreasing predation risk.[42,43] Highly arboreal primates such as red colobus and Diana monkeys show few morphological adaptations for terrestrial locomotion, yet opportunistically engage in terrestrial behaviors when predation risk is low, indicating that primarily arboreal primates engage in terrestriality when perceived risks are minimal.[43]

The threat posed by terrestrial predators is surely a strong disincentive for arboreal primates to descend to the ground. For example, the relative lack of terrestrial behavior in Sumatran orangutans compared to Bornean orangutans is likely due to the presence of tigers on Sumatra.[44] Similarly, a community of African forest monkeys on islands with few to no predators engage in higher rates of terrestriality than their mainland counterparts that are subject to regular predation.[45] Furthermore, spider monkey terrestrial behavior may be mostly limited by the threat of predation, as populations in Mexico and Central America use the ground more often than populations in South America, where predator densities are higher.[46] A recent study found similar results, with Panamanian capuchins on islands that lack mammalian predators showing a higher degree of terrestriality than mainland groups exposed to intact predator communities.[47]

3.4 | Food availability

Animal movement and habitat use have long been linked to the distribution and abundance of food resources. For arboreal primates, utilizing novel terrestrial resources may be advantageous. Indeed, it has been argued that the shift to terrestriality from an arboreal lifestyle among catarrhines was likely initiated to exploit terrestrial food resources (reviewed in McCrossin et al.[3]). Examples in extant primates are numerous. Black-and-white snub-nosed monkeys seasonally feed on terrestrial grasses during nutritionally lean seasons.[48,49] The relatively high rates of terrestriality in some populations are attributed to the lack of arboreal food resources, compared to populations in more productive habitats that are less terrestrial.[49] Ground use in semiterrestrial vervet monkeys was highest when normalized difference vegetation index (NDVI) values were low, potentially due to a relative increase in terrestrial food resources during these periods.[50] Among the great apes, the amount of time spent in the trees is correlated with the amount of fruit in the diet. For example, the more frugivorous lowland gorillas are more arboreal than their comparatively herbivorous mountain relatives.[51,52]

In the Neotropics, some monkeys have been observed regularly accessing terrestrial food resources as well. Bearded capuchins, for example, descend to the ground almost exclusively to forage on palm nuts, terrestrial bromeliads, and underground food storage organs.[53]

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Coimbra-Filho's titi monkeys increased terrestrial fruit and leaf consumption when arboreal resources were low.[54] However, some arboreal primates have expanded their diets to include terrestrial foods throughout the year, such as graminoids, forbs, and fungi, with no evidence of terrestriality being associated with seasonal fluctuations in arboreal food resources such as fruits and flowers.[55]

3.5 | Weather

Temperature is one of the most important environmental factors influencing animal behavior patterns.[56] Ambient air temperature can influence the terrestrial behaviors of primates as well. For example, air temperature was found to be the strongest predictor of terrestrial behaviors in both West African chimpanzees and Central African bonobos; both spent more time on the cooler forest floor when temperatures increased.[57] Similarly, temperature was the strongest predictor of terrestriality in a population of chimpanzees in Uganda, who used the ground more often during the hot afternoon hours.[58] In contrast, Southern bamboo lemurs spent more time foraging terrestrially during colder periods compared to warmer periods.[55]

Arboreal primates may also descend to the ground during dry seasons or periods of low rainfall. For example, primates will descend from the trees to access terrestrial sources of water during dry periods.[46,59] Capuchins living in seasonal habitats spend more time on the ground during dry seasons than wet seasons and are more terrestrial than capuchins inhabiting rainforest habitats (reviewed in Wright et al.[53]), suggesting terrestrial water sources are more important during periods of low rainfall. Additionally, terrestrial behaviors may be less common or absent in forest habitats that are seasonally flooded, perhaps due to decreased locomotor efficiency or to avoid standing water.

3.6 | Disturbance

Anthropogenic habitat alteration is often associated with significant changes to important ecological factors such as forest canopy connectivity, predator presence, and food availability.[60] These factors can influence primate terrestrial behaviors. For example, a six-fold increase in terrestrial behaviors was documented in a group of the highly arboreal lion-tailed macaques inhabiting a logged forest fragment as logging continued throughout the study period.[61] Similarly, terrestrial behavior in muriquis was most common in human-disturbed habitats.[62] Also, bale monkeys inhabiting forest fragments spent significantly more time on the ground (>33%) than groups in continuous forest (2%[63]). The exact mechanism connecting habitat disturbance and an increase in terrestrial behaviors among arboreal primates is surely case-specific but likely includes some of the factors detailed above. Given the influence of habitat disturbance on the terrestrial behaviors of primates, in some cases leading to extreme changes in terrestrial behavior, our analyses only consider terrestriality data for primates in habitats with little to no known anthropogenic disturbance.

3.7 | Sex differences

In mammals, females typically invest substantially more in reproduction than males, resulting in adaptations for the efficient utilization of nutritional resources—feeding success is vital for females' reproductive success. Hence, females are referred to as the "ecological sex" and are assumed to be more sensitive to ecological conditions than males.[64,65] Conversely, the typical mammalian male maximizes their fitness by prioritizing access to females over foraging efficiency. If female primates are more sensitive to ecological factors than males in general, we may expect this to be apparent in terrestrial behavior as well, with female primates being more or less terrestrial than their male counterparts in response to certain factors. Therefore, our analyses will primarily consider female primates in models of terrestriality but include a male-female averaged model for comparison.

4 | EXISTING DATA

To explore associations between primate terrestriality and primate characteristics and ecology, and to test the hypotheses of terrestriality detailed above, data concerning factors that may influence terrestriality in primates are needed. In addition to the terrestriality data collected, we also compiled data on primate characteristics and ecological attributes of their ranges.

4.1 | Potential correlates of terrestriality

We compiled data on primate characteristics and ecology from databases and published materials. These measures include body mass, percentage of fruit in the diet, along with the following attributes averaged across a taxon's range: forest height, canopy cover, the number of potential predator species, NDVI, maximum temperature, and average rainfall.

Primate body mass data were mostly reported as a male-female average, with fewer sources reporting female-only and/or male-only body mass. For this reason, we created two measures of body mass, male-female averaged mass and female-only mass. Due to the paucity of sex-specific dietary data, we used average percent fruit in the diet for both male and female adults.

The geographic range area for each taxon was determined using shapefiles available on the IUCN Spatial Data Download web page, accessed January 2022, only considering extant taxa, not including "possibly extant" range designations. When spatial range data were not available for species, we combined range data for its subspecies, yielding range data for 732 taxa. We used ranges to calculate ecological variables for primates. It should be

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noted that these range maps may provide only coarse representations of primate range areas, which may influence the calculations of our area-based metrics below. Despite this potential imprecision, we have little reason to suspect the range designations are systematically biased.

We collected forest characteristic data using the Global Forest Canopy Height 2019 data set,[66] a global raster of 30 m spatial resolution, extracting the median forest height for each primate's range. We collected canopy cover data by accessing the 2021 MODIS/Terra Vegetation Continuous Fields data set[67] using NASA Earthdata Search, isolating the "Percent Tree Cover" layer and extracting the median value for each taxon's range.

To create a metric of predator presence we accessed range data from the IUCN spatial database for Carnivora taxa. We retrieved carnivore body mass and terrestriality data from ADW and PanTHERIA databases, selecting species designated as terrestrial. Following Eppley et al.,[68] we considered only carnivores whose body masses constituted 25% or more of the mass of the focal primate. We then calculated the total number of carnivores with ranges overlapping each taxon's range.

To retrieve NDVI data, we used Google Earth Engine to access the Visible Infrared Imaging Radiometer Suite (VIIRS) raster imagery and calculated the median NDVI value for each pixel at 1 km resolution for the period between 2012 and 2022. We then extracted the median NDVI value across each taxon's range.

We collected weather data from WorldClim ver. 2.1,[69] extracting the maximum temperature (BIO1) and average rainfall (BIO12) bioclimatic variable raster layers, both at 1 km resolution. Using the same process for the raster layers above, we extracted the median value of bioclimatic variables across each taxon's range.

4.2 | Modeling approach

We analyzed all primate data using Bayesian models in R Statistical Software (ver. 4.2.1[70]) using the *brms* package.[71] We controlled for phylogenetic effects in all models by including a variance-covariance matrix derived from the 10kTrees primate phylogeny. Three models were built using the three terrestriality classifications. We ran each model twice; a female-only version and a sex-averaged version. We modeled the binary terrestriality outcome variable using a Bernoulli distribution and logit link-function, the ordinal outcome variable using a cumulative probability distribution and probit link-function, and the continuous outcome using a zero-inflated beta distribution and logit link-function. We used weakly informative priors for all models, using a normal distribution with an SD of 10 for the intercept and an SD of 1 for all predictors. We ran all models using 8000 iterations, with 3000 warmup iterations, over four Markov chain Monte Carlo chains, resulting in 20,000 total postwarm-up draws.

All models include the variables described above and the continental location of taxa (Americas, Africa, Madagascar, or Asia). We scaled and centered all numerical predictors to allow for a direct comparison of coefficient estimates. We used credible intervals (CI's) for posterior distributions of predictor coefficients to define reliable predictors, considering 90% CI's that did not overlap with zero, the default in *RStan* functions, to be reliable indicators of directionality for a predictor, and 50% CI's that did not overlap zero as weakly reliable predictors. We checked all predictors for issues with multicollinearity post model-fitting with most predictors being weakly correlated with one another, with variance inflation factors (VIF) below 5. The only exceptions were canopy cover in the continuous models showing moderate correlation (VIF = 6), and NDVI, forest height, and canopy cover showing moderate correlation (VIF = 5–8.5) in submodel two. However, no issues with model convergence were apparent.

4.3 | Results

4.3.1 | Full models

Binary models included 188 taxa in the female-only model (164 arboreal, 24 terrestrial) and 215 taxa in the sex-averaged model (190 arboreal, 25 terrestrial). In the female model, only body mass was a reliable, positive indicator of terrestriality. Three predictors were weakly reliable: NDVI (negative), maximum temperature (negative), and annual rainfall (negative). In the sex-averaged model, no predictors were reliable indicators of terrestriality, and two predictors were weakly reliable: body mass (positive) and NDVI (negative; Figure 3a).

Ordinal models included 163 taxa in the female-only model (129 arboreal, 16 semiterrestrial, 18 terrestrial), and 186 taxa in the sex-averaged model (150 arboreal, 17 semiterrestrial, 19 terrestrial). In the female model, two predictors were reliable indicators of terrestriality: body mass (positive) and NDVI (negative). Two predictors had weak associations with terrestriality: rainfall (negative) and the number of carnivores in range (negative). In the sex-averaged model, only NDVI was a reliable, negative indicator of terrestriality. Three predictors had weak associations with terrestriality: body mass (positive), rainfall (negative), and carnivores (negative; Figure 3b).

Continuous models included 113 taxa for the female-only model and 127 taxa for the sex-averaged model, with ground use ranging from 0% to 98.4% in both (Q1 = 0, med = 1.9, mean = 14.5, Q3 = 18.6). In the female model, two predictors had reliable effects on terrestriality: body mass (positive) and NDVI (negative). Three predictors were weakly reliable: percent fruit in diet (negative), forest height (positive), and rainfall (negative). In the sex-averaged model, only NDVI was a reliable, negative indicator of terrestriality. Three predictors were weakly reliable: body mass (positive), percent fruit in diet (negative), and forest height (positive; Figure 4).

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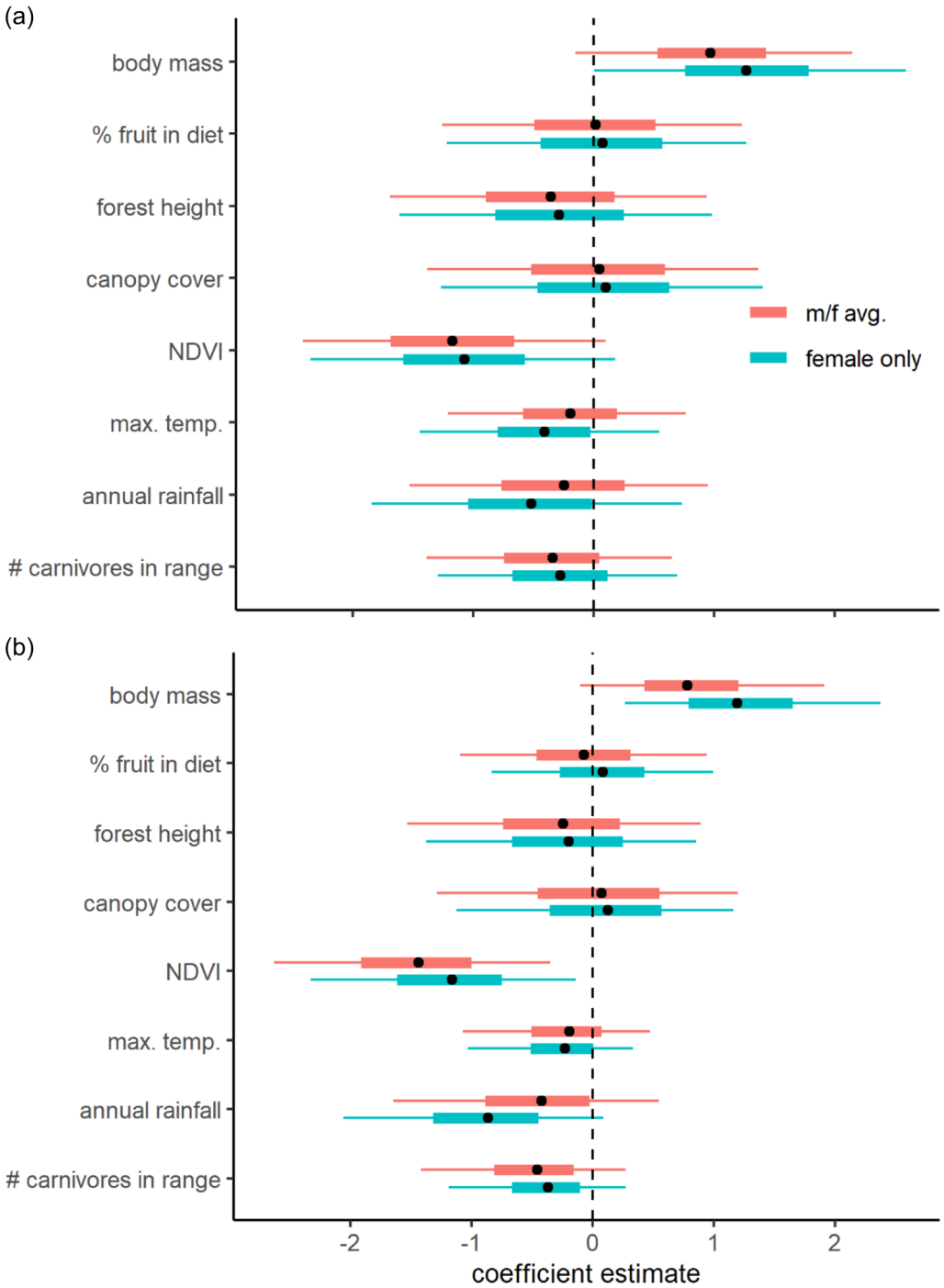

FIGURE 3 Coefficient plots showing results of the binary (a) and ordinal (b) models for the sex-averaged (m/f avg.) and female-only models. Plots depict coefficient estimates (black points), 50% Bayesian credible intervals (CI's; bars), and 90% CI's (lines).

4.3.2 | Submodels

Some authors have differentiated between primates that engage in terrestriality habitually and show clear adaptations for terrestrial locomotion (e.g., hominins, cercopithecines) and those that engage in terrestriality opportunistically and show no clear adaptations for moving on the ground (e.g., platyrrhines, lemurs). To test the effect of the variables included in the full models above on different groups of primates, we ran two submodels on subsets of taxa. Submodel one included only catarrhines. Submodel two included platyrrhines and lemurs. Both submodels use the continuous terrestriality measure as the outcome variable, using the same predictors as the full models.

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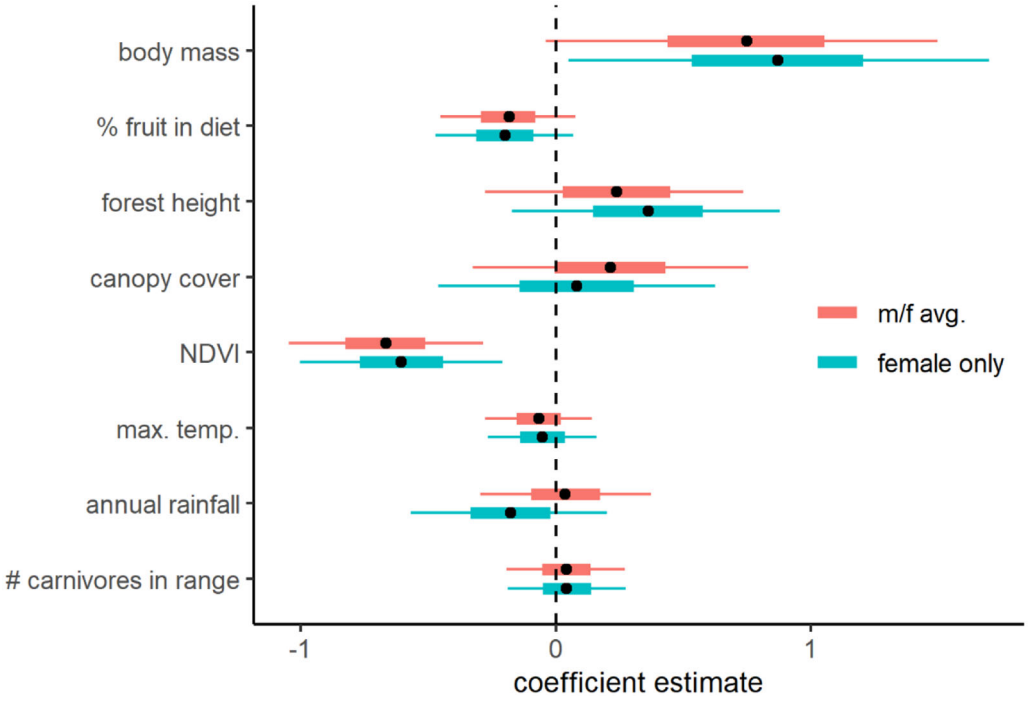

FIGURE 4 Coefficient plot showing results of the continuous model.

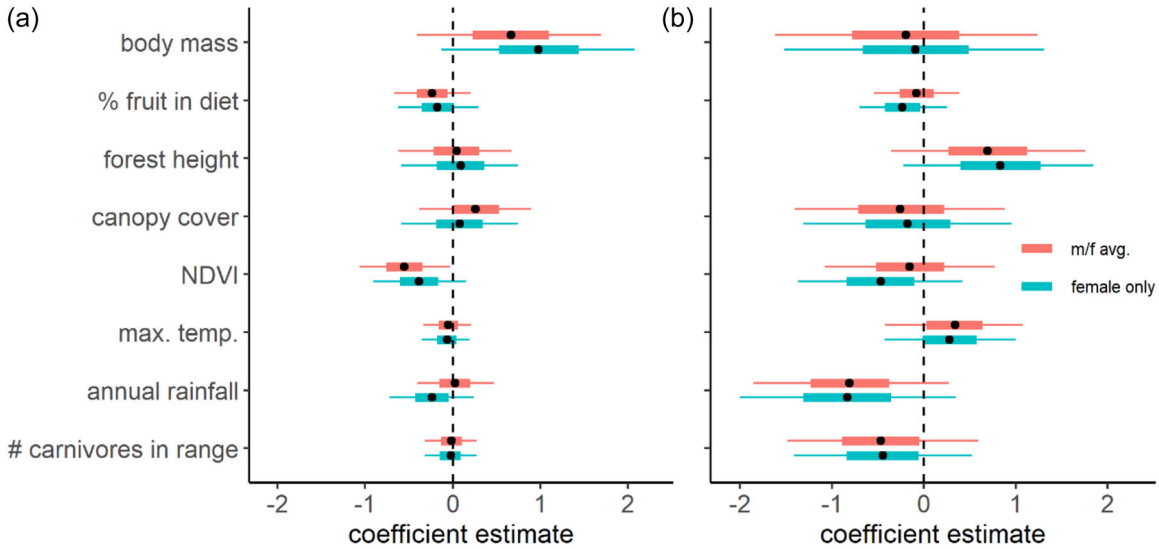

FIGURE 5 Coefficient plots showing model results for submodel one (a), including taxa from Cercopithecoidea and Hominoidea, and submodel two (b), including taxa from Platyrrhini and Lemuriformes.

Due to a lack of available data, we did not include Lorisiformes or Tarsiiformes in submodels.

Submodel one included 64 taxa for the female-only model and 68 taxa for the sex-averaged model. No predictors in the female model were reliable indicators of terrestriality. Three predictors were weakly reliable: body mass (positive), NDVI (negative), and annual rainfall (negative). In the sex-averaged model, NDVI is the only reliable, negative indicator of terrestriality. Two predictors were weakly reliable: body mass (positive), and percent fruit in diet (negative; Figure 5a).

Submodel two included 41 taxa for the female-only model and 46 taxa for the sex-averaged model. No predictors in either model were reliable indicators of terrestriality. In the female model, five predictors were weakly reliable: percent fruit in diet (negative), forest height (positive), NDVI (negative), annual rainfall (negative),

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and carnivores (negative). In the sex-averaged model, four predictors were weakly reliable: forest height (positive), temperature (positive), annual rainfall (negative), and number of carnivores in range (negative; Figure 5b).

Table 1 presents detailed results for all models listed above.

5 | DISCUSSION

5.1 | Body mass best predicts terrestriality

In many of our models, body mass was the most reliable predictor of ground use in primates. Based on the widespread connection between terrestriality and body mass in the primate literature and given the observation that some of the largest primates are highly terrestrial, this result is unsurprising. This lends support to hypothesis one, that primates will increasingly use the ground as body mass increases.

While body mass was the most reliable predictor of ground use in our full models, our analyses of subsets of primates found similar results for catarrhines (submodel one) but not for platyrrhines and lemurs (submodel two), indicating that body mass is more influential for the terrestrial behavior of catarrhines compared to other clades. This may be due to the larger body masses of catarrhines ($M = 11.2$ kg) compared to platyrrhines and lemurs ($M = 2.7$ kg) in our data set. However, body mass not being a reliable predictor in submodel two may reflect the substantially lower amount of variation in this variable (SD = 2.7 kg) compared to the variation in submodel one (SD = 8.1 kg), reducing our ability to detect a possible signal of body mass.

While increased body mass has long been associated with an increase in terrestriality in primates, few studies have explicitly tested this assumption. Previous work on single species found differences in ground use between adult and juvenile primates, citing differences in body mass (e.g. Zhu et al.,[39]). Interspecific comparisons of primate terrestriality are still rare, however. Possibly only one study has systematically reviewed the effect of body mass, among other variables, on terrestrial behavior across many species,[68] which modeled the terrestrial behavior of 47 diurnal platyrrhines and lemurs, finding a negative effect of body mass on terrestriality. This contrasts with our results from submodel two, finding no reliable effect of body mass in these clades.

The use of body mass in our models prevented the inclusion of other potentially informative predictors, as body mass is correlated with many mammalian characteristics such as home range, group size, body length, endocranial volume, and population density.[18] Indeed, we found a high correlation between primate body mass and some of these variables, and a moderate correlation with home range (Spearman's rank correlation coefficient [ρ] = 0.64) and group size ($\rho = 0.59$). The inclusion of home range and group size in our full models resulted in convergence issues and models with worse fit according to formal comparisons using Widely Applicable Information Criterion (WAIC), so we did not include these in our final analyses. Furthermore, we found a systematic bias in the availability of published home range and group size data. Taxa for which home range data are available are significantly heavier ($M = 6.73$ kg, SD = 15.19 kg) than those lacking home range data ($M = 3.35$ kg, SD = 4.93 kg; $t[303.48] = 3.20$, $p = 0.002$). Similarly, taxa for which group size data are available are significantly heavier ($M = 6.85$ kg, SD = 15.82 kg) than those lacking group size data ($M = 3.57$ kg, SD = 4.91 kg; $t[264.27] = 2.90$, $p = 0.004$). However, we found group size to be moderately correlated with primate ground use ($\rho = 0.69$), and in most models in which it was included it was a reliable positive predictor of terrestriality, with an effect size second only to that of body mass in our full models. Since both primate home range and group size have long been interpreted as responses to resource availability and an adaptation to predation pressure, respectively,[72] and not as a predictor of terrestriality per se, we did not formally consider either in the present analyses. Ultimately, these factors may have facilitated a shift to terrestriality in some primate lineages, perhaps in concert with an increase in body mass, based on their apparent covariation. Further work in this area is needed to unravel the complexities of the interactions between these primate characteristics.

5.2 | Mixed effects of food resources

Here, we use NDVI as a proxy for plant productivity, describing the plant biomass of an area, and by extension, the availability of potential primate foods. In our full models, NDVI was consistently a reliable negative indicator of terrestriality, indicating primates in ranges with lower average NDVI values are more terrestrial than primates in ranges with higher average NDVI values. This effect is slightly attenuated in our submodels, with NDVI only a reliable predictor of terrestriality in the sex-averaged submodel one. This decrease in effect strength may be due to fewer taxa being considered in submodels, possibly limiting our ability to reliably estimate the impact of NDVI in these groups. If NDVI is successfully describing arboreal food resources, then these results may lend some support for hypothesis four. However, NDVI may only be a useful proxy for green, photosynthetic material (e.g., leaves), and not for other arboreal food sources, namely fruit, and so may only be a useful indicator of food resources for folivorous primates. Alternatively, NDVI may only be capturing broad differences in primate habitat types. Habitats with a high density of photosynthetic material (i.e., leaf cover) such as tropical rainforests tend to have high NDVI values, while this value is lower in shrub and grassland habitats, where many terrestrial primates are found.

We found no reliable effect of frugivory on terrestriality, with only weakly negative associations seen in the continuous models and some submodels, providing only some evidence that more terrestrial primates may be less frugivorous. It is possible that the presence of frugivorous primates is a reasonable proxy for the variance of fruit availability over time, with their presence indicating high habitat fruit availability. If so, then this would lend some support to hypothesis four. However, the weak associations found suggest that the amount of fruit in the trees has little influence on primate terrestriality, or that the amount of fruit in the diet is not a suitable proxy for habitat fruit availability.

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TABLE 1 Details for all models used in the current analysis.

	Binary		Ordinal		Continuous		Submodel 1		Submodel 2	
	(m/f)	(fem)	(m/f)	(fem)	(m/f)	(fem)	(m/f)	(fem)	(m/f)	(fem)
Body mass (sex-averaged)	0.983 (0.702)		0.830 (0.624)		0.741 (0.467)		0.658 (0.640)		−0.200 (0.866)	
Body mass (female only)		1.276 (0.788)		1.240 (0.650)		0.869 (0.501)		0.982 (0.676)		−0.093 (0.862)
Percent fruit in diet	0.007 (0.759)	0.052 (0.760)	−0.076 (0.625)	0.081 (0.562)	−0.185 (0.160)	−0.200 (0.165)	−0.234 (0.266)	−0.170 (0.280)	−0.078 (0.281)	−0.232 (0.289)
Forest height	−0.366 (0.799)	−0.298 (0.794)	−0.271 (0.742)	−0.220 (0.685)	0.236 (0.309)	0.360 (0.319)	0.040 (0.391)	0.086 (0.404)	0.695 (0.642)	0.828 (0.631)
Canopy cover	0.028 (0.834)	0.081 (0.819)	0.028 (0.765)	0.088 (0.700)	0.214 (0.326)	0.082 (0.332)	0.262 (0.389)	0.077 (0.402)	−0.253 (0.696)	−0.177 (0.685)
NDVI	−1.167 (0.767)	−1.076 (0.764)	−1.464 (0.692)	−1.192 (0.668)	−0.666 (0.231)	−0.605 (0.242)	−0.552 (0.313)	−0.381 (0.324)	−0.153 (0.560)	−0.473 (0.544)
Maximum temperature	−0.203 (0.604)	−0.422 (0.602)	−0.234 (0.476)	−0.274 (0.421)	−0.067 (0.127)	−0.052 (0.129)	−0.054 (0.169)	−0.070 (0.171)	0.333 (0.453)	0.280 (0.437)
Annual rainfall	−0.263 (0.758)	−0.532 (0.781)	−0.471 (0.668)	−0.906 (0.657)	0.038 (0.203)	−0.179 (0.233)	0.026 (0.266)	−0.236 (0.290)	−0.800 (0.642)	−0.826 (0.716)
Carnivores in range (sex-averaged)	−0.352 (0.618)		−0.502 (0.520)		0.041 (0.141)		−0.016 (0.182)		−0.465 (0.629)	
Carnivores in range (female only)		−0.285 (0.610)		−0.399 (0.452)		0.042 (0.141)		−0.025 (0.180)		−0.446 (0.591)
Num.Obs.	215	188	186	163	127	113	68	64	46	41
R^2	0.782	0.79	0.896	0.876	0.355	0.391	0.435	0.464	0.562	0.542

Note: Coefficient estimates are shown with standard errors in parentheses. Models were fit twice, including predictors that used sex-averaged data (m/f) or female-only data (fem). The number of taxa included in each model (Num.Obs.) and R-squared (R^2) values are also included.

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5.3 | Mixed effects of weather variables

While temperature was not consistently a reliable or weakly reliable predictor of terrestriality, average annual rainfall was weakly negatively associated with terrestriality in most of our models, lending little support to hypothesis five. This relationship may be driven by the fact that some of the driest primate habitats (<1000 mm annual rainfall) are inhabited by the most terrestrial species (>70% ground use) in our analyses (i.e., *Theropithecus gelada*, *Papio* spp., and *Macaaca sylvanus*), while primates inhabiting the wettest habitats (>2500 mm annual rainfall) are nearly all predominantly arboreal species (<30% ground use). However, rainfall has a much larger effect on terrestriality in platyrrhines and lemurs (submodel two) than it does for catarrhines (submodel one). In fact, rainfall has the largest effect size of any predictor in submodel two, suggesting this variable is more influential for these clades. This is in agreement with a review of Neotropical primates finding many species in dry woodland habitats exhibiting more terrestrial behavior than their rainforest relatives, and finding rainforest species are more terrestrial during dry seasons, citing the need to access terrestrial water sources.[53]

Without more detailed data pertaining to annual or seasonal variations in terrestrial behavior, we are limited to broad descriptions of the influence of weather variables on primate terrestriality. This may be why the results of submodel two mostly contrast with the work of Eppley et al.,[68] who used monthly measures of temperature and rainfall and found a reliable positive effect of temperature.

5.4 | No reliable effect of forest structure

Canopy cover and forest height were not reliable predictors of terrestriality in any of our models. However, forest height had a weak positive association with terrestriality in our continuous models and in submodel two, with it having the second largest effect size in the latter, suggesting forest height may be somewhat influential for terrestriality in platyrrhines and lemurs but not for catarrhines. These findings provide weak support for hypothesis two.

A potential explanation for this effect, however tenuous, involves competition and niche separation. Tall, highly stratified forests harbor more primate species and at higher densities than shorter forests,[73] potentially intensifying resource competition. One mode of niche separation is to utilize all available forest strata, including the forest floor. For instance, macaques exhibit greater terrestriality in part as a response to competition with potential competitors.[74] If competition within primate communities is driving species to be more terrestrial in taller forests, it remains unclear why this isn't apparent among catarrhines in our analyses. Previous work has argued that competition is less prevalent among Asian primate communities.[75] If this is true for terrestrial behaviors as well, perhaps this partially explains why catarrhine terrestriality is not impacted by forest height.

Alternatively, these simple metrics of forest structure may not adequately describe the structural complexity of the forest that is meaningful to primates. More complex aspects of forest structure, such as vertical heterogeneity and horizontal connectivity, may have a more important impact on primate movement and terrestrial behavior (e.g., Davis & Asner[76]).

5.5 | No reliable effect of carnivore presence

We found no reliable effect of potential predators on primate terrestriality in our full models, and a weak negative influence in our ordinal models and submodel two, lending little support to hypothesis three. This may be due to morphological differences between the taxa in submodels one and two. The more opportunistic ground users in submodel two generally lack obvious adaptations for ground use and may be more susceptible to terrestrial predation compared to the more terrestrially fluent catarrhines. Alternatively, the metric of predation we use may not accurately capture the effect that predators have on primates. Our metric represents the number of carnivore species in a primate's range, based on relative carnivore body mass, but does not account for predator density, predation threat, or the effectiveness of predator strategies. The presence of one highly effective predator, for example, is likely more influential to primate ground use than the presence of multiple carnivores that seldom prey on primates. A more effective metric may better reflect the true threat predators pose to primates or primates' perceived risk of predation.

5.6 | Females as the ecological sex

Our analyses lend some support to the notion of females being the so-called ecological sex. Coefficient estimates for predictors in the female-only models show mostly minor differences compared to sex-averaged models. This could be due to an actual lack of sex differences in the influence of ecological variables on terrestriality. Alternatively, this could be attributed to the relative scarcity of sex-specific data available, particularly for female primates, which may have affected our model results. Our female models consistently considered fewer taxa than our sex-averaged models. Also, our female models only contain two female-specific predictors; body mass and number of predators in a taxon's range, which is calculated using body mass. Data pertaining to female-specific ground use and percentage of fruit in the diet were not sufficiently available for our analyses. However, since female primates and their infants are likely targeted more by predators than males are, we may expect a general trend of females being less terrestrial than males.[49,77]

5.7 | Limitations

Understanding primate terrestrial behavior remains a work in progress, reliant on the availability of detailed, long-term ground use data for species across a diversity of phylogenetic groups and habitats. While we were able to gather relatively lower resolution terrestriality data (binary and ordinal) for many taxa (65% and 54%, respectively), high-resolution (continuous) data were available for

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only approximately 30% of all primate taxa. Continuous data were especially limited for some families, including Aotidae (0/14 taxa with continuous data), Cheirogaleidae (6/40), and Callitrichidae (15/70).

Our sample sizes constrained the number of predictors we could incorporate in our models, and inclusion of some potentially informative predictors (e.g., locomotor anatomy) were beyond the scope of this paper. Therefore, our models surely do not capture the full complexity of terrestrial behavior in primates and do not consider several potential alternative explanations. For example, sympatric species may be highly similar in respect to the predictors included in our models (i.e., body mass, predation pressure, weather, and forest structure metrics of ranges) but engage in terrestriality to vastly different degrees, such as in sympatric chimpanzees and baboons, indicating that other factors, potentially antifeedant-tolerance or the spatial distribution of food, may be underlying this difference.[5]

Because of the relatively low number of taxa considered in some of our models, and the number of predictors we used, adding interaction terms was not feasible. Models that included interactions either failed to converge or produced unstable coefficient estimates. Adding interactions would require a much larger data set than we have compiled here, as they require considerably more data to estimate than main effects (e.g., interactions of the same size as main effects require four times as much data to estimate reliably[78]). The inability to include interactions here is unfortunate, as it is unlikely that any one of our predictors is solely responsible for the presence of terrestriality in primates. For example, an interaction between body size and weather variables such as temperature likely influences primate terrestrial behavior. Smaller animals, given their high surface area to volume ratios, tend to be more impacted by extreme temperatures[79] and may be more likely to utilize behavioral strategies of thermoregulation to maintain appropriate body temperatures,[80] such as descending to the cooler forest floor.

Many variables included in our models are averaged values, such as average canopy cover across a species' range and average species terrestriality estimates. Using averages has drawbacks, such as decreased precision in the characterization of certain attributes of a species. In the case of terrestriality, using average estimates of the percent time on the ground from sometimes limited data sources representing few, or perhaps only one, population(s), may not accurately reflect the terrestrial behavior of the species, and certainly masks any variability of this behavior within species. Also, using the average body mass between female and male adults of taxa, for example, may not accurately reflect the true influence of body size of either sex. This will be especially pronounced for taxa that are highly sexually dimorphic. Large differences in body mass between males and females of sexually dimorphic primates can influence substrate use,[81] including degrees of terrestriality.[82] However, species that are highly sexually dimorphic do not always show large differences between the sexes in degree of terrestriality.[51,53]

Using averages in the present work allowed us to maximize the number of taxa included in the final data set, as many taxa do not have detailed data concerning variables of interest for both sexes. For the same reason, we only consider terrestriality, body mass, and dietary composition data for adult primates, and do not explicitly include data concerning juveniles. However, we acknowledge that these characteristics can vary widely between sex and age classes for certain primates. For example, adult and juvenile capuchin males spend more time on the ground (18% and 8% of observations, respectively) than both adult and juvenile females (2% and <1%[83]).

Our analyses only consider the influence of environmental/ecological factors and some species characteristics on primate terrestriality. Previous work has considered the influence of intrinsic characteristics on primate substrate use, specifically terrestrial behavior. For example, Fragaszy[83] suggested that the difference in ground use between capuchin males and females is most likely due to differences in psychological characteristics affecting the response to perceived risk, with males, more so than females, showing a greater willingness to engage in risky behaviors, such as spending more time on the ground. Similarly, previous work has estimated the influence of other primate characteristics, such as locomotor type, on terrestriality, finding no reliable effect in platyrrhines or lemurs.[68] While further investigation may yield interesting results in this area, systematically collecting data pertaining to species locomotor anatomy is beyond the scope of this paper.

It is important to note that we did not consider study duration or degree of habituation in the present study, a limitation that might particularly affect the continuous terrestriality measure. One reason for this is that these data are not available in the databases we used, and thus for a substantial portion of the taxa we considered in our models. Ensuring terrestriality data accurately represent this behavior in primates is important, as shorter studies may not adequately reflect ground use for species over different seasons, especially for species that rarely engage in terrestriality. Likewise, degree of habituation is not available in databases and is seldom reported in publications but is likely to influence in-situ observations of terrestrial behavior for many primates.[84]

5.8 | Future directions

We conducted our analyses at the taxon scale, using averaged species characteristic metrics and ecological variables across species ranges and over time. Future work on the topic of primate terrestrial behavior would benefit from focusing on smaller spatial and temporal scales, such as terrestriality at the population or individual level, and ecological variables at annual or monthly scales. Indeed, such work has recently been undertaken for a subset of diurnal and arboreal primate populations.[68] More detailed work on primate populations and communities would be helpful to expand such efforts to include African and Asian primates, and nocturnal species as well.

Focusing on smaller scales of analysis also permits researchers to explore the intricacies of highly complex ecosystems like tropical rainforests. One such example is microenvironmental differences. Vertical climate gradients in forests can be more pronounced than climate gradients along latitudes and elevations.[85] This is especially true for tropical forests, which have a particularly complex three-

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dimensional structure, providing a diversity of microenvironments and niches.[86] Fruit quantity and quality also tend to be vertically stratified in tropical forests. The forest canopy, compared to lower strata, generally has a greater abundance and density of larger fruit that contain higher water and sugar content.[87,88] How these microenvironmental differences influence primate terrestrial behavior has yet to be explored.

Traditional methods of collecting primate locomotor behavior data consist of direct in situ observations, which may influence the frequency and duration of terrestrial behaviors of primates, even those that are accustomed to human presence (e.g., Souza-Alves et al.[89]). Camera trap surveying is an ideal approach to observe terrestrial behaviors in primates, providing researchers with the means to capture rare behaviors without the risk of disrupting their natural expressions, and should be deployed whenever possible for this purpose.

6 | CONCLUSIONS

We describe the distribution of primate terrestriality both geographically and phylogenetically, finding terrestriality to be most common in Africa and Asia, and within the Papionini. Our ancestral state reconstructions suggest terrestriality evolved separately in at least two primate lineages around the middle Miocene. However, more detailed analyses are needed using higher resolution terrestriality data for more taxa to improve the precision of estimates of when and in which lineages terrestriality evolved.

Our full models show that primate body mass and average NDVI across a primate's range are the best predictors of terrestriality across the primate order. However, when considering subsets of taxa, this pattern holds for Catarrhini, but not for Platyrrhini and Lemuriformes. For the latter, our models show that average forest height and rainfall across primate ranges are the best, however weak, predictors of terrestriality. Our analyses of subsets of taxa are limited by the availability of terrestriality data, but suggest that terrestriality in platyrrhines and lemurs, groups that conspicuously lack fully terrestrial species, is influenced more by ecological/environmental factors compared to the more dominant influence of body mass in catarrhines.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data and supplementary materials are available at: https://github.com/Estrada-Gene/primate_terrestriality.

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